Chapter 3 Notes

3.1

In the proof of Theorem 3.1, here is why the class of all transitive natural numbers is a set. We can show that it is a subset of ω, and is hence a set by A_2. For suppose x is a member of the class of all transitive natural numbers. Then obviously x is also a natural number, hence $x \in \omega$, thus completing the proof.

Note that here is why the only natural numbers allowed for by Definition 2.2 are the ones defined right before Definition 1.1. By Definition 2.2, a natural number is a member of every inductive set. Now, consider the following inductive set: $\emptyset$ is an element, and if x is an element, then so is x^+. Hence every natural number is a member of this set. But this set is just the set of all sets that were defined in the way just before Definition 1.1.

Exercise 3.1

Suppose that A is a class of natural numbers such that for every natural number n, if A contains all numbers less than n, then A contains n as well. Now, we can prove by induction that A contains all natural numbers. Consider the property P meaning A contains all numbers natural numbers less than or equal to n. Now, there are no natural numbers less than 0. Thus the statement that A contains all natural numbers less than 0 is vacuously true; hence A contains 0. Thus, because 0 is the only natural number less than or equal to 0, P holds for 0. Now suppose that P holds for some n, that is, A contains all natural numbers less than or equal to n. But then A contains all numbers less than $n+1$, meaning (by the definition of A) A also contains $n+1$. Thus, by induction, A contains all natural numbers.

Exercise 3.2

Consider a natural number n.

(a): Suppose that $x \in n$. Now, it cannot be that $n = \emptyset$. Thus n is non-empty, hence of the form $(n-1)^+$. Thus either $x \in n-1$ or $x = n-1$. Now, $\cup n$ is just the set of all elements of all elements of $n-1$ together with all elements of $n-1$. Now suppose $x = n-1$. Then $x \notin \cup n$. For suppose it were. Then it would either be an element of $n-1$ or it would be an element of an element of $n-1$. In the first case, we would have $n-1 \in n-1$, which is ruled out by Theorem 3.2. In the second case, there would be some y such that $x \in y$ and $y \in n-1$. But then, by the transitivity of $n-1$ (by Theorem 3.1), $x \in n-1$, that is, $n-1 \in n-1$, which is again ruled out by Theorem 3.2. Thus $n-1 \notin \cup n$, meaning the statement is not necessarily true.

It should be noted that all elements other than $n-1$ of n are members of $\cup n$. For all elements of n are either elements of $n-1$ or equal to $n-1$. Thus, if they are not equal to $n-1$, the element is an element of $n-1$. But, because $n-1 \in n$, then all elements of $n-1$ are elements of $\cup n$.

(b): Suppose that $x \in \cup n$. Thus either $x \in n-1$ or x is an element of some element of $n-1$. In the first case, because $n-1 \in n$, $x \in n$ (by the transitivity of n). In the second case, because $n-1$ is transitive, it follows that $x \in n-1$. But then, because $n-1 \in n$ and n is transitive, $x \in n$. Thus, in either case, $x \in n$, and thus $\cup n \subseteq n$. Hence the statement is necessarily true.

(c): By part (b) of this problem, it follows that $x \in \cup n$ implies $x \in n$. But $n^+ = n \cup \{n\}$. Thus all elements of n are in n^+, meaning $x \in n^+$. On the other hand, suppose that $x \in n^+$. Then it could be that $x = n$. But then, if $x \in \cup n$, we

would have either n being an element of an element of $n-1$, or $n \in n-1$. In the first case, the transitivity of $n-1$ would show that $n \in n-1$. Thus we would have, in either case, $n \in n-1$, which is impossible. For if $n \in n-1$, then, because $n-1 \in n$, $n-1 \in n-1$ by the transitivity of $n-1$, which is impossible (or, if $n-1 = \emptyset$, it could have no members). Hence it is not necessarily the case that $n^+ \subseteq \cup n$. Thus $\cup n = n^+$ does not necessarily hold.

Similar reasoning as to the above also shows that $n-1$ cannot be an element of $\cup n$. It should be noted that $n-1$ and n are the only members of n^+ that are not members of $\cup n$. For any other element of n^+ is an element of n but not equal to $n-1$. But $n = n-1 \cup \{n-1\}$. Hence it would be an element of $n-1$, and because $n-1$ is an element of n, it would also be an element of $\cup n$.

(d): On the one hand, suppose that $x \in \cup n^+$. Then either x is an element of an element of n, or x is an element of n. In the first case, because n is transitive, $x \in n$. The second case means $x \in n$. Thus, in any case, $x \in n$. On the other hand, suppose that $x \in n$. Then $x \in \cup n^+$, as $n^+ = n \cup \{n\}$. Hence $\cup n^+ = n$.

(e): By part (b) of this problem, $\cup n \subseteq n$ holds. But by the answer to part (a) of this problem, it does not necessarily follow that $n \subseteq \cup n$. Thus it is not necessarily true that $\cup n = n$.

Exercise 3.3

(a): On the one hand, suppose that $x \in \cup\omega$. Then x is an element of a natural number. Thus either $x \in n$ or $x = n$. Now we prove the following statement: every element of every natural number is a natural number. The base case vacuously holds, as $\emptyset$ has no elements. Now suppose the statement holds for some n. Then $n^+ = n \cup \{n\}$. Hence, if $y \in n^+$, either $y \in n$ or $y = n$. In the former case, by the induction assumption, it follows that y is a natural number, and in the latter case, y would be a natural number (as n is a natural number). Thus, in any case, y is a natural number, therefore completing the proof. As a consequence of having proven this statement, it follows that x is a natural number, and hence $x \in \omega$. On the other hand, suppose that $x \in \omega$. Thus, for some n, $x = n$. Now, all elements of n^+ are in $\cup\omega$. But n is an element of n^+. Thus $n \in \cup\omega$, and hence $x \in \cup\omega$. Thus we have proven that $\cup\omega = \omega$.

(b): On the one hand, suppose that $x \in \cup\omega^+$. Now, $\omega^+ = \omega \cup \{\omega\}$. Thus either x is an element of a natural number or x is a natural number. By the results of part (a) of this problem, it follows that if x is an element of a natural number, then x is a natural number. Thus x is a natural number, that is, $x \in \omega$. Now suppose that $x \in \omega$. Then $x \in \cup\omega^+$, as x would be an element of ω, which is an element of ω^+. Thus we have proven that $\cup\omega^+ = \omega$.

Note that a generalization cannot be made of this problem. Define $\{\omega_0, \omega_1, \omega_2, ..., \omega_n, ...\}$ in the following way: $\omega_0 = \omega$ and $\omega_{n+1} = \omega_n^+$. Then it does not follow that $\cup\omega_n = \omega$ for all n. For consider $n = 2$. This is ω_1^+, that is,

$$\omega_1 \cup \{\omega_1\} = (\omega \cup \{\omega\}) \cup \{(\omega \cup \{\omega\})\}$$

Thus $\omega \in \cup\omega_1^+$. But $\omega \notin \omega$, meaning $\cup\omega_2 \neq \omega$. Indeed, it is a general fact that, for $n \geq 2$, $\cup\omega_n \neq \omega$. To prove this, we can just prove that, for all $n \geq 2$, $\omega \in \omega_n$. We have just proven the base case. If the result holds for some n, then $\omega \in \omega_{n+1}$, as $\omega_{n+1} = \omega_n \cup \{\omega_n\}$, and it was assumed that $\omega \in \omega_n$. Thus, for every $n \geq 2$, $\omega \in \omega_n$. Now we can prove the statement that, for $n \geq 2$, $\cup\omega_n \neq \omega$. We proved the base case for $n = 2$ above. Now consider some $n > 2$. Then $\omega_n = \omega_{n-1} \cup \{\omega_{n-1}\}$ and $\cup\omega_n$ consists of all elements of all elements of ω_{n-1} and all elements of ω_{n-1}. Because ω is an element of ω_{n-1}, it follows that $\omega \in \cup\omega_n$, and thus, because $\omega \notin \omega$, $\cup\omega_n \neq \omega$.

Section 4

Notes

Note that g-induction only works on minimally inductive sets for the following reason: If A was not minimally inductive under g, we would have some element x of A such that x is not a term of the g-sequence (see the section below titled Exercise 4.1) $\emptyset, g(\emptyset), g(g(\emptyset)), ...$ But the proof by g-induction only proves the result for the terms of the g-sequence started by $\emptyset$, meaning we would not necessarily know that the result held for x.

Note that (1) of Lemma 4.6 holds because 0 is just $\emptyset$, and $\emptyset$ obviously is a subclass of y for every y.

Theorem 4.16

The proof of Theorem 4.16 seems incorrect to me. At a point in the proof, we rely upon Theorem 4.12. But that assumes L is a bounded *subset* of A. However he assumes that $\emptyset \notin A$ and shows that $\emptyset \in L$, meaning L is not a subset of A, contrary to the claim that it was. (In fact, I think we would need to show that L is a subset of M, as M is the one which is minimally inductive under g, but Smullyan does not make this clear)

I think that what Smullyan was trying to say was: Let L be the class of all elements of x *of* M such that x is a proper subclass of all elements of A. Obviously, this means that L is a subset of M, and then, when we show that L is bounded by A, we really can use Theorem 4.12. Also, when we assume that $\emptyset \notin A$, we can still conclude that $\emptyset \in L$, as $\emptyset \in M$ by the assumption that M is minimally inductive under g.

Exercise 4.1

We suppose that N is closed under g, where g is progressing, and that b is an element of N and that no proper sublcass of N containing b is closed under g. In order to prove the desired results, we can first prove (1) of Theorem 4.17. We are given that N is closed under g. Call a sequence of the form $x, g(x), g(g(x)), g(g(g(x))), ...$ a $g-sequence$ (and call it a subset of a set A if all elements of the sequence are elements of A). Label x the first term of the g-sequence, $g(x)$ the second term, and, in general, if y is the nth term of the g-sequence, then $g(y)$ is the $n+1$th term of the g-sequence (and we say that n is the *index* of the nth term of the sequence). We first show that N is just the g-sequence $b, g(b), g(g(b)), ...$ Now, it is obvious that the g-sequence started by b is a subset of N (because N is closed under g, and b is in N). Then suppose that there was some element $c \in N$ outside of the g-sequence started by b. Then the g-sequence $c, g(x), g(g(c)), ...$ would be a subset of A. But then $b, g(b), g(g(b)), ...$ would be a proper subset of N containing b which was closed under g, contrary to the hypothesis. Hence the only elements in N are the elements of the g-sequence started by b. Having shown this, recall that g is a progressive function. Now consider any x, y in N. If $x = y$, then $y \subseteq x$. Now suppose $x \neq y$. Let x be the mth term of the g-sequence and y the nth term. Thus $m \neq n$. On the one hand, suppose that $m < n$. An obvious induction argument shows that $x \subseteq y$. Now either $g(x) = y$ or not. In either case, $g(x) \subseteq y$ is obvious. On the other hand, suppose that $n < m$. Then the same argument shows that $y \subseteq x$, thus completing the proof of (1).

Here is the "obvious inductive argument" mentioned above (I will refer to this as Lemma 1). We prove that, for all n, all mth terms where $m < n$ are subsets of the nth term of the g-sequence. The base case is obvious (as there are no terms of index less than 1). Now suppose the result holds for the nth term x. Then all terms of index m where $m < n$ are subsets of x. But then, because g is progressing, x is a subset of $g(x)$ (the $n+1$th term), and because all the terms of index m where $m < n$ are subsets of x, it follows that all terms of index m where $m < n+1$ are subsets of $g(x)$, completing the proof.

We now prove (2) of the problem. We prove by induction on n that every non-empty subset of N that is bounded by the nth term of N (that is, the nth term of the g-sequence $b, g(b), g(g(b)), ...$) has a greatest element. The base case is when $n = 0$, that is, when the term is b. Now consider some bounded subset c of N which is bounded by b. Thus $c \subseteq N$ and, for all $d \in c$, $d \subseteq b$. But, by Lemma 1, b is a subset of all elements of N. Thus b is a subset of all elements $d \in c$. But $d \subseteq b$. Hence $b = d$ for all elements $d \in c$, meaning c is just the set $\{b\}$. Then, of course, b is the greatest element of c, therefore proving the base case. Now suppose the hypothesis holds for some n. That is, every non-empty subset of N bounded by the nth term x of N has a greatest element. Now, the $n+1$th term of N is just $g(x)$. Consider any non-empty subset B of N bounded by $g(x)$. If $g(x) \in B$, then of course $g(x)$ is the greatest element of B. So suppose that $g(x) \notin B$. Then every element of B is a proper subset of $g(x)$. Consider any element y of B. Then $y \subset g(x)$, meaning $g(x) \subseteq y$ doesn't hold, and therefore $y \subseteq x$ holds (because either $g(x) \subseteq y$ or $y \subseteq x$ by (1) of this problem). But this means that B is bounded by x; hence, by the induction assumption, it has a greatest element.

I was not entirely sure what (3) was trying to prove: I interpreted it to mean that, if x is the greatest element of N, then it is a fixed point of g. Suppose that x is the greatest element of N. Then, for all elements y of N, $y \subseteq x$. Now, we know that $g(x) \in N$. Because x is the greatest element of N, $g(x) \subseteq x$. But also, because g is progressing, $x \subseteq g(x)$. Thus $g(x) = x$, meaning x is a fixed point of g.

Now we prove (4) of the problem. Because we have shown that $g(x) \subseteq y$ or $y \subseteq x$ for all x, y in N, it follows that $x \subseteq y$ or $y \subseteq x$ (as $x \subseteq g(x)$). Thus N is a nest. It remains to be shown that every non-empty subclass of N has a least element. So consider some non-empty subclass A of N. Because N is just the set of all terms of the g-sequence $b, g(b), g(g(b)), ...$, it follows that A contains only members of that sequence. Hence A contains some term x whose index (in the g-sequence) n is less than or equal to the indices of all the other elements of A. By Lemma 1, it follows that x is a subset of all terms with an index greater than or equal to n, and hence x is a subset of all other elements of A, meaning it is the

least element of A.

Because b starts the g-sequence $b, g(b), g(g(b)), ...$, which makes up the entirety of N, Lemma 1 above shows that b is the least element of N (is a subset of all elements of N). This proves (5) of the problem.

The reason that this generalizes some of the earlier results of the section is that it takes away the requirenment that N be minimally inductive under g. For example, Theorem 4.7 holds now when N is closed under g, where g is progressing, and that b is an element of N and that no propert subclass of N containing b is closed. That is, we do not need to know that N is minimally inductive under g.

Exericse 4.2

(I did not think that this proof requires A to be a set)
Suppose that A is inductive under g and that A is a set. Thus we know
(1) $\emptyset \in A$,
(2) for every x in A, the set $g(x)$ is in A.
Consider the g-sequence $\emptyset, g(\emptyset), g(g(\emptyset)), ...$ Because A is inductive under g, that g-sequence is a subset of A. But clearly, that g-sequence is minimally inductive under g. Hence there is some subset of A that is minimally inductive under g.

While we cannot show that every g-inductive class is a set (as Smullyan remarked, our axioms are not yet powerful enough to show that every denumerable class is a set), we can show that every element of every g-inductive class is a set. Recall that the definition of a function relied upon the definition of relations, which in turn relied upon the notion of an ordered pair $\langle x, y\rangle$, which was only defined when x and y are sets. Thus x can only be in the domain of a function if x is a set. Now consider any g-inductive class A, and consider any element x in A. Then we know that $g(x) \in A$, and thus, because x is in the domain of g, x is a set. If it happens that A is finite (which could occur if, for instance, $\emptyset$ was a fixed point of g and A only contained $\emptyset$), A would be a set as a consequence of Exercise 6.1. Indeed, there are non-denumerable g-inductive classes, for instance, the class of all sets, which is g-inductive under the power function.

Section 5

Exercise 5.1

Suppose that the natural number n is in A and that for any number x in A, x^+ is also in A. We can prove that A contains every number $\geq n$ by g-induction. In particular, consider the property P: If $m \geq n$, then $m \in A$. We show that this prpoerty holds for all numbers m. Either $\emptyset = n$ or $\emptyset < n$. In the first case, P holds, while in the second case, P vacuously holds. Thus P holds for $\emptyset$. Now suppose that P holds for x. There are two cases: either $x^+ < n$, $x^+ = n$, or $x^+ > n$. In the first case, P vacuously holds. In the second case, because we assumed $n \in A$, P holds. In the last case, by the induction assumption that $x \in A$, $x^+ \in A$, and thus P holds for x^+. Thus, in any case, P holds for x^+, therefore completing the induction.

Exercise 5.2

We suppose f is a function from ω into A.

(a): We suppose that, for all n, $f(n) \subseteq f(n^+)$. We can prove that, for all n and m, if $n \leq m$ then $f(n) \subseteq f(m)$ by induction on m. In particular, we show that, for all numbers m, for all numbers $n \leq m$, $f(n) \subseteq f(m)$. The base case is when $m = \emptyset$. The only natural number $\leq \emptyset$ is $\emptyset$, which is a subset of itself, meaning the base case holds. Now suppose that the hypothesis holds for some m. Thus, for all $n \leq m$, $f(n) \subseteq f(m)$. Now, we are given that $f(m) \subseteq f(m^+)$. But this means that, for all number $n \leq m$, $f(n) \subseteq f(m^+)$. Thus all numbers $n \leq m^+$ are such that $f(n) \subseteq f(m^+)$, therefore completing the induction.

(b): We prove that, for all n, if $f(n) \subset f(n^+)$, then for all n and m, if $n < m$, then $f(n) < f(m)$. The base case is when $m = \emptyset$. In this case, there are no natural numbers less than $\emptyset$, so the property vacuously holds. No suppose it holds

for some m. Thus, for all $n < m$, $f(n) \subset f(m)$. Now, we are given that $f(m) < f(m^+)$. But this means that, for all $n < m$, $f(n) \subset f(m^+)$, thus completing the induction.

Section 6

Exercise 6.1

(1): We prove that every finite class is a set. This can be done using induction on the number of elements n in the class. The base case is for the class $\emptyset$, which is a set by A_3. Now suppose the result holds for some n. Consider some class S of size $n+1$. Pick some arbitrary element a of S. By the induction assumption, $S - \{a\}$ is a set. Also, $S = \{a\} \cup (S - \{a\})$. But $\{a\} \cup (S - \{a\}) = \cup\{\{a\}, (S - \{a\})\}$, which is a set by A_5 (because $\{\{a\}, (S - \{a\})\}$ is a set by A_4), thus proving S is a set.

(2): We prove that every subset of a finite set is finite. This can be done using induction on the number of elements n in the set. The base case is for $\emptyset$, whose only subset is $\emptyset$, which is finite. Thus the base case holds. Now assume the result holds for some n. Consider some finite set of S size $n+1$. Consider any subset F of S. Either it is a subset of a set of size n, or it is not. If it is, the induction assumption shows that it is finite. Otherwise, the only way that it could not be a subset of a set of size n would be if F was S itself (because S only has $n+1$ elements), meaning F would be finite, therefore completing the proof.

(3): We prove that every non-empty finite set of natural numbers has a greatest element. This can be done using induction on the natural numbers. The base case is for $\{\emptyset\}$, whose only element, and therefore greatest element, is $\emptyset$. Now assume that the result holds for some n, that is, n has a greatest element. Now consider $n+1$. It is just $n \cup \{n\}$. Thus its elements are either elements of n, or n itself. But we know that all elements of n are subsets of n (by results in section 5). Thus all elements of $n+1$ are subsets of n, therefore completing the proof.

Exercise 6.2

We prove that the union of any two finite sets is finite. An equivalent thing to prove is that the statement that the union of a set S with n elements and any other finite set is finite, holds for all n. This can be proven using complete induction on n. The base case, when $n = 0$, is when $S = \emptyset$. Now consider any other finite set A with m elements. But, because $\emptyset \cup A = A$ holds for all sets A, and because A was assumed to be finite, $\emptyset \cup A$ is also finite. Thus the base case is proven. Now suppose the result holds for all numbers less than n, that is, for any set S having less than n elements, $S \bigcup A$ has a finite amount of elements for all finite sets A. Now consider some set B having n elements. Pick some arbitrary element a of B. Then $B = (B - \{a\}) \cup \{a\}$. Now consider any finite set C. Then $C \cup B = C \cup ((B - \{a\}) \cup \{a\})$. But, in general, for any sets X, Y and Z, $X \cup (Y \cup Z) = (X \cup Y) \cup Z$. Thus $C \cup B = (C \cup (B - \{a\})) \cup \{a\}$. But $B - \{a\}$ has $n-1$ elements. Thus, by the induction assumption, $C \cup (B - \{a\})$ is finite. Also, $\{a\}$ is finite. Thus we know by the induction assumption that, because $C \cup (B - \{a\})$ has been shown to be finite, $(C \cup (B - \{a\})) \cup \{a\}$ is finite, that is, $C \cup B$ is finite, which completes the induction.

Exercise 6.3

We prove that the union of any finite set S of finite sets is finite. We prove this by induction on the size n of S. In the base case, S has only one element a, and its union is just all of the elements of a, which is finite, as a was assumed to be finite. Now suppose the result holds for some n. Consider some set S of size $n+1$ of finite sets. Let its elements be $a_1, a_2, ..., a_n, a_{n+1}$. Let $S_1 = \{a_1, a_2, ..., a_n\}$. Now, $\cup S = (\cup S_1) \cup a_{n+1}$ (as can easily be shown). By the induction assumption, $\cup S_1$ is finite. Then, by Exercise 6.2, $(\cup S_1) \cup a_{n+1}$ is also finite. Hence $\cup S$ is finite, therefore completing the proof.

Exercise 6.4

Suppose that M is minimally under g where g is progressing. We prove that for every x in M, the set of all $y \in M$ such that $y \subseteq x$ is finite. This can be done by using g-induction on M. For $\emptyset$, it only has a subset of $\emptyset$, meaning the set of all $y \in M$ such that $y \subseteq x$ is finite. Now suppose the result holds for some x. That is, the set S of all $y \in M$ such that $y \subseteq x$ is finite. Now consider $g(x)$. Because g is progressing, $x \subseteq g(x)$. Thus all elements of S are subsets of $g(x)$. Now suppose there was some $y \in M$ such that $y \subseteq g(x)$, but y is not a member of S (that is, y is not a subset of x). Thus, because M is a nest (as was shown in section 4), $x \subseteq y$. Now, we know that $y \subseteq g(x)$. Thus $x \subseteq y \subseteq g(x)$, meaning, by Theorem 4.10 part (1), either $y = x$ or $y = g(x)$. Thus the elements of the set of all $y \in M$ such that $y \subseteq g(x)$ are the elements of S, and the sets x and $g(x)$. But $x \in S$. Thus the set of all $y \in M$ such that $y \subseteq g(x)$ is just the set S together with $g(x)$, meaning it is also a finite set (because S is finite).

We can also show the result using g-sequences: Suppose that M is minimally under g where g is progressing. We prove that for every x in M, the set of all $y \in M$ such that $y \subseteq x$ is finite. Now, M is just the set of all members of the g-sequence $\emptyset, g(\emptyset), g(g(\emptyset)), ...$ We prove that for all n, the nth term in the above g-sequence satisfies the desired conditions. This shows that all elements of M satisfy the desired conditions. Consider the nth term in the sequence x. Because g is progressing, we know that every mth term of the sequence where $m \leq n$ is a subset of the nth term. These elements of course form a finite set, one with $n - 1$ elements (call this set S). Now, there are no other elements of M which are subsets of x. For suppose there was one, call it y. Then it would have an index greater than n (because all the terms with an index $\leq n$ were already shown to be subsets of x). But then, because g is progressing, $x \subseteq y$. But also $y \subseteq x$, meaning $x = y$, proving that y is in S. Thus the set of all $y \in M$ such that $y \subseteq x$ is just S, which is finite, therefore completing the proof.

Exercise 6.5

Suppose M is minimally inductive under g, where g is progressing. Suppose also that M contains an element x such that $x = g(x)$. By Theorem 4.14, we then know that x is the greatest element of M, that is, for all $y \in M$, $y \subseteq M$. But by Exercise 6.4, the set of all $y \in M$ such that $y \subseteq x$ is finite. Then, because every element of M is a subset of x, it follows that M is finite.

We can also prove this using g-sequences: Suppose M is minimally inductive under g, where g is progressing. Suppose also that M contains an element x such that $x = g(x)$. We show that M is finite. We know that M is just the set of all the members in the g-sequence $\emptyset, g(\emptyset), g(g(\emptyset)), ...$ Suppose the first element x of M for which $x = g(x)$ has an index n in that g-sequence. Then, as can easily be shown, for all terms y with an index greater than n in the sequence, $y = x$. Thus the only elements of M are the first n terms of the g-sequence, meaning M is finite, in particular, a set with n elements.

Section 7

Exercise 7.1

We can prove the equivalent statement that, if a set S of natural numbers is not finite, then it is denumerable. Note that, because ω is the set of all natural numbers, $S \subseteq \omega$. We can make a 1-1 correspondence between S and ω as follows: pick some element a_0 of S and pair it with 0. Then pick another element a_1 of S and pair it with 1, and, in general, we pair n with some element a_n of S, where a_n is disitinct from all a_m where $m < n$. If there was any n for which it could not be paired to an a_n, this would mean that S only had $n - 1$ elements, contrary to the assumption that it was infinite. If there was ever some a_n for which no n could be found, this would contradict the fact that $S \subseteq \omega$. Thus this really is a 1-1 correspondence.

We can also show that every non-denumerable set has a denumerable proper subset. For consider any non-denumerable set A. We can show that some proper subset of A can be put into a 1-1 correspondence with ω in the following way: Pick some element a_0 of A, and pair it with 0. Then pick another element a_1 of A, and pair it with 1, and in general, for every n, we pick a new element a_n of n to match with n. If for any n we could not pick a new element of A to match with it, that would mean that A would only have $n - 1$ elements, contrary to the assumption that it was infinite. Thus, for every n, we can pick an element of A to match with it, meaning we have a denumerable subset S of A consisting of the elements $a_0, a_1, ..., a_n, ...$ If this was the entirety of A, then A itself would be denumerable, contrary to the assumption

that it was not. Thus S is a proper subset of A, therefore completing the proof.

Note the following incorrect proof: Having shown the above result, we can now prove that any infinite subset of a denumerable set is denumerable. For consider some denumerable set A. Suppose it had a non-denumerable subset B. By the above result, there is a proper subset C of B which is denumerable. Because $B \subseteq A$ and $C \subset B$, $C \subset A$. Thus we would have A being in a 1-1 correspondence with C, a proper subset of A, which is impossible. Hence A can have no non-denumerable subset. This is of course incorrect because it is not true that A cannot be in a 1-1 correspondence with a proper subset of itself. For instance, ω can be placed in a 1-1 correspondence with the set of all even numbers.

Note the following incorrect proof that any denumerable set can be put into a 1-1 correspondence with any non-denumerable set: Consider any denumerable set A and any non-denumerable set B. Arrange the elements of A in the order $a_0, a_1, ..., a_n, ...$ If for any n, a_n did not have a matching element in B, then B would be finite, a contradiction. Also, if we ever encountered an element in B for which we could pair no element of A with, this would mean A would be finite, a contradiction. The error in this proof is that we of course can reach an element in B for which we could pair no element of A with (and this does not mean that A is finite; it would only mean so if the statement was:"if we ever encountered some *nth element* in B for which we could pair no element of A with" This would then mean that A was finite). Of course, this element in B that we could not pair up with A would come after pairing off a denumerable amount of elements from B, meaning we would already know that A was infinite (and thus there is no supposed contradiction).

We can prove the fact that no denumerable subset can have a non-denumerable subset in a different way. First, define a set S to be *greater* than a set F if S cannot be put into a 1-1 correspondence with F, but some proper subset of S can be put into a 1-1 correspondence with F. If S is greater than F, we denote this by $S > F$, and say that F is less than S, and denote this by $F < S$. We first show some facts about sets being greater than one another, and then the desired result comes easily.

First we show that, for any sets S and F, either $S < F$ or $F < S$, or S and F can be put into a 1-1 correspondence. Suppose that S and F cannot be put into a 1-1 correspondence. Pick some element a_1 of S, and b_1 of F, and pair them together. Then pick some new element a_2 of S, and some new element b_2 of F, and so on. Becaus S and F cannot be put into a 1-1 correspondence, we will at some point either reach some element of F for which no element of S can be found, in which case $F > S$ (as the the elements paired up with S up until that point represent a subset of F, and is in a 1-1 correspondence with S), or we reach some element of S for which no element of F can be found, in which case $S > F$.

Lemma: *For all sets S, no subset of S can be greater than S.*
For suppose some subset F was greater than S. Then S and F could not be put into a 1-1 correspondence, but some proper subset of F could be put into a 1-1 correspondence with S. We can show that F and S can, in fact, be put into a 1-1 correspondence, in the following way: If there was ever some element of F for which we could not find any element of S to pair it with, this would contradict the fact that F was a subset of S. If there was ever some element of S for which we could not find any element of F to pair it with, this contradicts the fact that some proper subset of F could be put into a 1-1 correspondence with S. Thus S and F can be put into a 1-1 correspondence, a contradiction, therefore completing the proof.

After this, we can prove the desired result about denumerable sets (that no denumerable set can have a non-denumerable subset). For suppose we had some denuemrable set A having a non-denumerable subset B. We have already shown that some proper subset of every non-denumerable subset is denumerable. Hence a proper subset of B is denumerable (can be put into a 1-1 correspondence with A). But, by definition, B cannot be put into a 1-1 correspondence with A. Thus $B > A$, which contradicts the Lemma, therefore completing the proof.

We now show that, for any sets S and F, it is not the case that both $S > F$ and $F > S$ hold. For suppose the hypothesis did not hold. Then we would have S and F not in a 1-1 correspondence, and some proper subset A of S being in a 1-1 correspondence with F, and some proper subset B of F being in a 1-1 correspondence with S. Then we would have A, a proper subset of S, being put into a 1-1 correspondence with F, a set which is greater than S. Thus A, a subset of S, is greater than S, which is impossible by the Lemma, therefore completing the proof.

The following Lemma will be useful: **Union Lemma**: *For any sets A, B, C, and D, if A and D can be placed in a 1-1 correspondence, and B and C can be placed in a 1-1 correspondence, and A and B contain no elements in common, and C and D contain no elements in common, then $A \cup B$ can be placed in a 1-1 correspondence with $C \cup D$.*

Proof: Suppose we have sets A, B, C, and D in the relation described in the statement of the Lemma. First, we pair off the elements of A in the set $A \cup B$ with the elements of C in the set $C \cup D$, which can be done, as A and C can be placed in a 1-1 correspondence. Also, this does not use any elements of B, as A and B were assumed to be disjoint. Then we can do the same thing for B and D, therefor showing we can pair up all elements of $A \cup B$ with $C \cup D$, meaning $A \cup B$

can be placed in a 1-1 correspondence with $C \cup D$.

We now show that any infinite set can be put into a 1-1 correspondence with a proper subset of itself. We first prove this for denumerable sets, and then the case for non-denumerable sets easily follows. Consider any denumerable set S. Thus the elements of S can be arranged in a denumerable sequence $a_1, a_2, ..., a_n, ...$ Now consider the set $S - \{a_1\}$, which is a proper subset of S. We can show that it can be placed in a 1-1 correspondence with S as follows: Note that the elements of $S - \{a_1\}$ are $b_1, b_2, ..., b_{n+1}, ...$, where $b_1 = a_2$, $b_2 = a_3$, and, in general, $b_n = a_{n+1}$. We pair up a_1 with b_1, a_2 with b_2, and, in general, a_n with b_n, thus showing that there is a 1-1 correspondence between S and $S - \{a_1\}$, a proper subset of S.

Now that we have shown the denumerable case, consider some non-denumerable set S. We have already shown that S contains some denumerable subset D. Now pick some arbitrary element a of D. Now, $D \cup (S - D) = S$. Consider the set $(D - \{a\}) \cup (S - D)$ (which is just the set $S - \{a\}$), which is a proper subset of S. We have already shown that $D - \{a\}$ can be placed into a 1-1 correspondence with D. Then, because $D - \{a\}$ can be placed in a 1-1 correspondence with D, $S - D$ can be placed in a 1-1 correspondence with $S - D$, D and $S - D$ are disjoint, and $D - \{a\}$ and $S - D$ are disjoint, the Union Lemma shows that $D \cup (S - D)$ can be placed in a 1-1 correspondence with $(D - \{a\}) \cup (S - D)$, that is, S can be placed in a 1-1 correspondence with a proper subset of itself.

Now we prove that, for any element a of any infinite set S, there is a denumerable subset of S containing a. We have already shown that S must contain some denumerable set D. Either D contains a or it doesn't. If it does, we are done. If it doesn't, pick some arbitrary element e of D. Then $(D - \{e\}) \cup \{a\}$ is the desired set, as we can show that it is denumerable. We know that the elements of D can be placed in a denumerable sequence $a_0, a_1, ..., a_n, ...$ If $a_0 \neq e$, then we pair 0 off with a_0, and if $a_0 = e$, then we pair 0 off with a. If $a_1 \neq e$, then we pair 1 off with a_1, and if $a_1 = e$, then we pair 1 off with a. And, in general, if $a_n \neq E$, then we pair 1 off with a_n, and if $a_n = e$, then we pair n off with a, thus showing that $(D - \{e\}) \cup \{a\}$ is denumerable.

Now we prove that the union of any denumerable set and any finite set is denumerable. For consider some finite set F and some denumerable set D. Then the elements of D can be arranged in the sequence $a_1, a_2, ..., a_n, ...$ Also, the set G of all elements of F which are not already elements of D is finite, say its elements are in the sequence $b_1, b_2, ..., b_m$. Then $D \cup F = D \cup G = \{b_1, b_2, ..., b_m, a_1, a_2, ..., a_n, ...\}$. We pair off b_1 with the $a_1 \in S$, b_2 with the $a_2 \in S$, and, for all $i \leq m$, we pair off b_i with the $a_i \in S$. Then, we pair off the a_1 in $D \cup G$ with the a_{m+1} in D, the a_2 in $D \cup G$ with the a_{m+2} in D, and, in general, for all $i > m$, we pair off the $a_i \in D \cup G$ with the $a_{m+i} \in D$, thus showing that $D \cup G$ can be placed in a 1-1 correspondence with D.

Now we prove that the union of any non-denumerable set S (thus proving the result for all infintie sets) and any finite set can be placed in a 1-1 correspondence with S. For consider some non-denumerable set S and some finite set F. Let G be the set of all elements in F not in S. Now, we know that S contains some denumerable subset D. By the preceeding result, $D \cup G$ is denumerable, that is, $D \cup G$ can be placed in a 1-1 correspondence with D. Then, because $S - D$ can be placed in a 1-1 correspondence with $S - D$, and $D \cup G$ can be placed in a 1-1 correspondence with D, and $D \cup G$ and $S - D$ are disjoint (because G and S are disjoint), and D and $S - D$ are disjoint, the Union Lemma shows that $(D \cup G) \cup (S - D)$, that is, $S \cup G$, can be placed in a 1-1 correspondence with $D \cup (S - D)$, that is, S, therefore completing the proof.

Section 8

Exercise 8.1

Suppose that A is a non-empty set of natural numbers having no greatest element. We prove that A is denumerable. We have already shown (in Exercise 7.1) that A is either finite or denumerable. It suffices to show that every non-empty finite set of natural numbers has a greatest element, which would show that A must be denumerable. We prove this by induction on the size n of the finite set S. The base case is when $n = 1$. In this case, $S = \{m\}$ for some natural number m, meaning the greatest element of S is just m. Now suppose the result holds for some n, that is, any finite set of natural numbers of size n has a greatest element. Now consider any finite set F of natural numbers S having $n + 1$ elements. Pick some arbitrary element a of F. Then $S - \{a\}$ has n elements. By the induction assumption, it has a greatest element g. Now, either $g \subseteq a$ or $a \subseteq g$. In the first case, because g is the greatest element of $S - \{a\}$, all elements of $S - \{a\}$ are subsets of a. But the only elements of S are those of $S - \{a\}$ and a. Thus a is the greatest element of S. In the second case, because the only elements of S are those of $S - \{a\}$ and a, and because all of these are subsets of g, g is the greastest element of S. Thus, in any case, S has a greatest element, meaning, by induction, all non-empty finite sets of natural numbers have a greatest element. Thus, because A does not have a greatest element, it follows that A is denumerable.

Exercise 8.2

I was unsure what Smullyan meant by a "unique" function. What would a *non*-unique function look like?

We first show that for every n, there is a function f_n defined on the set of integers from 0 to n such that $f_n(0) = c$ and for each number $i < n$, $f_n(i^+) = g(f_n(i))$. We prove this by induction on n. The base case is when $n = 0$. Then we just take f_0 to be the ordered pair $\langle 0, c\rangle$. Now suppose the result holds for some n, that is, $f_n(0) = c$ and for all $i < n$, $f_n(i^+) = g(f_n(i))$. Then take f_{n+1} to be the set of all ordered pairs defined for f_n as well as the ordered pair $\langle n^+, g(f_n(n))\rangle$. Next we show that for each n, $f_n \subseteq f_{n^+}$, that is, for every number $k \leq n$, $f_n(k) = f_{n^+}(k)$. Now, $f_n(0) = f_{n^+}(0)$ follows by definition. Then suppose that $f_n(k) = f_{n^+}(k)$ for some $k < n$. Then $f_n(k^+) = g(f_n(k)) = g(f_{n^+}(k))$ and $f_{n^+}(k^+) = g(f_{n^+}(k))$, meaning $f_n(k^+) = f_{n^+}(k^+)$. Hence, for all $k \leq n$, $f_n(k) = f_{n^+}(k)$. Then we take f to be the the set of all ordered pairs $\langle n, f_n(n)\rangle$ (for all n). Lastly, we need to show that $f(0) = c$ and $f(n^+) = g(f(n))$. The first part holds as $f(0) = f_0(0) = c$. The second part also holds, as $f(n^+) = f_{n^+}(n^+) = g(f_{n^+}(n)) = g(f_n(n)) = g(f(n))$. Thus the proof is complete.

The proof above is fine, though I think there is a simpler way to prove the theorem. Consider any function g from A into A and any set $c \in A$. Then consider the g-sequence started by c: $c, g(c), g(g(c)), ...$ It is defined by the recursive rule $a_0 = c$ and $a_{n+1} = g(a_n)$. Then take the function f to be the set of all ordered pairs $\langle n, a_n\rangle$ (for all n). We need to show show that $f(0) = c$ and $f(n^+) = g(f(n))$. The first part holds, as $f(0) = a_0 = c$. The second part holds, as $f(n^+) = a_{n^+} = g(a_n) = g(f(n))$.

Exercise 8.3

Note that, up until this point, Smullyan has yet to define what a triadic function is. I will assume the definition goes as follows: First, we define what an n-ary relation is. Define V^n in the following way: $V^2 = V \times V$, $V^3 = (V \times V) \times V$, and $V^{n+1} = V^n \times V$ for $n \geq 2$. Then an n-ary relation is just a subclass of V^n. Now, let us recall the definition for an n-tuple: $\langle x_1, ..., x_n, x_{n+1}\rangle = \langle\langle x_1, ..., x_n\rangle, x_{n+1}\rangle$ when $n > 2$, and when $n = 2$, it is just $\langle x_1, x_2\rangle$. We can show by induction that an n-ary relation is just a class of n-tuples. The base case, when $n = 2$, is obvious. Now suppose the result holds for some n. Then $V^{n+1} = V^n \times V$, that is, all the elements in V^{n+1} are of the form $\langle x, y\rangle$ where $x \in V^n$. But it was assumed that all elements in V^n were n tuples. Thus all elements in V^{n+1} are of the form $\langle\langle x_1, ..., x_n\rangle, x_{n+1}\rangle = \langle x_1, x_2, ..., x_n, x_{n+1}\rangle$ (which holds by definition). Thus V^{n+1} is simply a class of $n+1$-tuples, completing the induction. Then we say that the domain of an n-ary relation R (for $n > 2$) is the class of all $n-1$-tuples $\langle x_1, ..., x_{n-1}\rangle$ such that $\langle\langle x_1, ..., x_{n-1}\rangle, y\rangle \in R$ for at least one y. By the range of R is meant the class of all y such that $\langle\langle x_1, ..., x_{n-1}\rangle, y\rangle \in R$ for at least one $n-1$-tuple $\langle x_1, ..., x_{n-1}\rangle$. Then a function of n (for $n > 2$) variables is a relation R such that for any $n-1$-tuple $\langle x_1, ..., x_{n-1}\rangle$ there is at most one element y such that $\langle\langle x_1, ..., x_{n-1}\rangle, y\rangle \in R$.

Now having defined functions of n variables, we can move on to some proofs. Consider any class A and any element c in A and any function $g(x, y)$ from $\omega \times A$ into A. We wish to show that there is some function f from ω into A such that $f(0) = c$ and for any number n, $f(n^+) = g(n, f(n))$. We first show by induction on n that for every n, there is a function f_n defined on the set of numbers from 0 to n such that $f_n(0) = c$ and for each number $i < n$, $f_n(i^+) = g(i, f_n(i))$. The base case is for $n = 0$, in which case we just take this to be the ordered pair $\langle 0, c\rangle$. Then suppose that the result holds for some n, that is, f_n is such that $f_n(0) = c$ and for all $i < n$, $f_n(i^+) = g(i, f_n(i))$. Then take f_{n^+} to be all the ordered pairs that f_n is defined on, as well as the ordered pair $\langle n^+, g(n, f_n(n))\rangle$ Now we show that for each n, $f_n \subseteq f_{n^+}$, that is, for every number $k \leq n$, $f_n(k) = f_{n^+}(k)$. This is true, as $f_n(0) = f_{n^+}(0)$ is obvious, and suppose that for some $k < n$, $f_n(k) = f_{n^+}(k)$. Then $f_n(k^+) = g(k, f_n(k))$ and $f_{n^+}(k) = g(k, f_{n^+}(k)) = g(k, f_n(k))$, that is, $f_n(k^+) = f_{n^+}(k^+)$, therefore showing the statement is true. Then we take f to be the set of all ordered pairs $\langle n, f_n(n)\rangle$. We need to show that $f(0) = c$ and $f(n^+) = g(n, f(n))$. The first part holds, as $f(0) = f_0(0) = c$. The second part holds, as $f(n^+) = f_{n^+}(n^+) = g(n, f_{n^+}(n)) = g(n, f_n(n)) = g(n, f(n))$.

This is a strengthening of Theorem 8.1, as Theorem 8.1 can be derived from this result in the following way: For consider some function g from A into A and any set $c \in A$. Take $h(x, y)$ to be the set of all triples $\langle x, y, g(y)\rangle$ (where $x \in \omega$ and $y \in A$). Then, by Exercise 8.3, there is some function f from numbers to elements of A such that $f(0) = c$ and for any number n, $f(n^+) = h(n, f(n)) = g(f(n))$, which satisfies the conditions of Theorem 8.1.

Exercise 8.4

(a): By Theorem 8.1, taking g to be the successor function, for any natural number n, there is some function f from numbers to numbers such that

(1) $f(0) = n$;
(2) for every n, $f(n^+) = f(n)^+$
Let the function satisfying the above for $n = 0$ be labeled f_0, the function satisfying the above for $n = 1$ be labeled f_1, and, in general, the function satisfying the above for $n = m$ be labeled f_m. Thus we have a denumerable sequence of functions $f_0, f_1, ..., f_m, ...$ such that for each i, $f_i(0) = i$ and for every n, $f_i(n^+) = f_i(n)^+$. Then define $A(x, y)$ to be the set of all triples $\langle m, n, f_m(n)\rangle$ (where $m \in \omega$ and $n \in \omega$). We need to show that for any natural numbers x and y, $A(x, 0) = x$ and $A(x, y^+) = A(x, y)^+$. The first part holds, as $A(x, 0) = f_x(0) = x$. The second part also holds, as $A(x, y^+) = f_x(y^+) = f_x(y)^+ = A(x, y)^+$.

(b): In order to prove this, we first prove the following lemma (for any class A):
Lemma: *For any function g from $A \times A$ into A and any set $c \in A$, there is a function f from $\omega \times \omega$ to elements of A such that:*
(1) for every m, $f(m, 0) = c$;
(2) for all m, n, $f(m, n^+) = g(m, f(m, n))$.
Indeed, the proof of the Lemma can easily be altered to find a fuction f such that (2) is changed to:
(2') for all m, n, $f(m, n^+) = g(f(m, n), m)$. Smullyan's desired multiplication function does not seem correct to me. For then we would have:

$$M(0, 2) = A(M(0, 1), 1) = A(A(M(0, 0), 0), 1) = A(0, 1) = 1$$

which is obviously not what we want.

This can be proven in the following way. First we show that for every n, there is a function $f_n(x, y)$ defined on all x and all $y \leq n$ (where $x \in \omega$ and $y \in \omega$) such that $f_n(x, 0) = c$ and for each number $i < n$, $f_n(x, i^+) = g(x, f_n(x, i))$. We prove this by induction on n. The base case is when $n = 0$. Then we just take f_0 to be the set of all triples $\langle x, 0, c\rangle$. Now suppose the result holds for some n, that is, $f_n(x, 0) = c$ holds for all x, and for all $i < n$, $f_n(x, i^+) = g(x, f(x, i))$ holds for all x. Then take f_{n^+} to be the set of all triples that f_n was defined on, as well as the triples $\langle x, n^+, g(x, f_n(x, n))\rangle$ for all x. Next we show that for each n, $f_n \subseteq f_{n^+}$, that is, for every number $k \leq n$, $f_n(x, k) = f_{n^+}(x, k)$ holds for all numbers x. Now, $f_n(x, 0) = f_{n^+}(x, 0)$ by definition. Then suppose that $f_n(x, k) = f_{n^+}(x, k)$ holds for all x for some $k < n$. Then $f_n(x, k^+) = g(x, f_n(x, k)) = g(x, f_{n^+}(x, k))$ and $f_{n^+}(x, k^+) = g(x, f_{n^+}(x, k))$, that is, $f_n(x, k^+) = f_{n^+}(x, k^+)$. Thus, by induction, $f_n(x, k) = f_{n^+}(x, k)$ holds for all x and all $k \leq n$. Then take f to be the set of all triples $\langle x, n, f_n(x, n)\rangle$ (for all x and n). We need to show that $f(m, 0) = 0$ for all m, and that $f(m, n^+) = g(m, f(m, n))$ for all m and n. The first part holds, as $f(m, 0) = f_0(m, 0) = c$ for all m. The second part holds, as $f(m, n^+) = f_{n^+}(m, n^+) = g(x, f_{n^+}(m, n)) = g(m, f_n(m, n)) = g(m, f(m, n))$. Thus f is the desired function.

Having proven the Lemma, the multiplication function comes easy: By the Lemma, there is some function M (from $\omega \times \omega$ into ω) such that:
(1) $M(x, 0) = 0$ holds for all x;
(2) for every x, y, $M(x, y^+) = A(x, M(x, y))$.
M is the desired multiplication function.

Having obtained the addition and multiplication functions, we can now prove that they are commutative. First we show this for the addition function $A(x, y)$.

It suffices to prove the following statement holds for all x: $A(x, y) = A(y, x)$ holds for all y. We prove this by induction on x. The base case is when $x = 0$. We know that $A(y, 0) = y$ for all y. We need to show that $A(0, y) = y$ for all y. The base case, when $y = 0$, holds, as $A(0, 0) = 0$. Now suppose the result holds for some y, that is, $A(0, y) = y$. Then $A(0, y^+) = A(0, y)^+ = y^+$. Thus, by induction, $A(0, y) = y$ for all y, meaning $A(y, 0) = A(0, y)$ for all y. Now suppose that the result holds for some x, that is, $A(x, y) = A(y, x)$ holds for all y. We need to show that $A(x^+, y) = A(y, x^+)$ holds for all y. The base case, when $y = 0$, holds, as it was already shown that $A(z, 0) = A(0, z)$ holds for all z. Now suppose the result holds for some y, that is, $A(x^+, y) = A(y, x^+)$. Then

$$A(x^+, y^+) = A(x^+, y)^+ = A(y, x^+)^+ = A(y, x)^{++} = A(x, y)^{++}$$

and

$$A(y^+, x^+) = A(y^+, x)^+ = A(x, y^+)^+ = A(x, y)^{++}$$

and thus $A(x^+, y^+) = A(y^+, x^+)$. Hence, by induction, for all x, $A(x, y) = A(y, x)$ holds for all y.

Before proving the commutativity of multiplication, we first prove the associativity of addition. That is, we prove that $A(x, A(y, z)) = A(A(x, y), z)$. It suffices to show that, for all x, $A(x, A(y, z)) = A(A(x, y), z)$ holds for all y and z. The base case is when $x = 0$. We know that

$$A(0, A(y, z)) = A(A(y, z), 0) = A(y, z)$$

and

$$A(A(0,y),z) = A(A(y,0),z) = A(y,z)$$

meaning $A(0,A(y,z)) = A(A(0,y),z)$. Now suppose the result holds for some x, that is, $A(x,A(y,z)) = A(A(x,y),z)$ for all y and z. Then

$$A(x^+,A(y,z)) = A(A(y,z),x^+) = A(A(y,z),x)^+ = A(x,A(y,z))^+$$

and

$$A(A(x^+,y),z) = A(A(y,x^+),z) = A(A(y,x)^+,z) = A(z,A(y,x)^+) = A(z,A(y,x))^+$$
$$= A(A(y,x),z)^+ = A(A(x,y),z)^+ = A(x,A(y,z))^+$$

Hence, by induction, for all x, $A(x,A(y,z)) = A(A(x,y),z)$ holds for all y and z.

Now we show that the multiplication function is commutative. It suffices to prove the following statement holds for all x: $M(x,y) = M(y,x)$ holds for all y. We prove this by induction on x. The base case is when $x = 0$. We know that $M(y,0) = 0$ holds for all y. We need to show that $M(0,y) = 0$ for all y. The base case, when $y = 0$, holds, as $M(0,0) = 0$. Now suppose the result holds for some y, that is, $M(0,y) = 0$. Then $M(0,y^+) = A(0,M(0,y)) = A(0,0) = 0$. Thus $M(0,y) = 0$ holds for all y, meaning $M(y,0) = M(0,y)$ holds for all y. Now suppose the result holds for some x, that is $M(x,y) = M(y,x)$ holds for all y. We need to show that $M(x^+,y) = M(y,x^+)$ holds for all y. The base case, when $y = 0$, holds, as it was already shown that $M(z,0) = M(0,z)$ holds for all z. Now suppose the result holds for some y, that is, $M(x^+,y) = M(y,x^+)$. Note that $A(x^+,y) = A(y^+,x)$, as

$$A(x^+,y) = A(y,x^+) = A(y,x)^+ = A(x,y)^+$$

and

$$A(y^+,x) = A(x,y^+) = A(x,y)^+$$

Then

$$M(x^+,y^+) = A(x^+,M(x^+,y)) = A(x^+,M(y,x^+)) = A(x^+,A(y,M(y,x))) = A(x^+,A(y,M(x,y)))$$
$$= A(A(x^+,y),M(x,y)) = A(A(x,y)^+,M(x,y))$$

and

$$M(y^+,x^+) = A(y^+,M(y^+,x)) = A(y^+,M(x,y^+)) = A(y^+,A(x,M(x,y)))$$
$$= A(A(y^+,x),M(x,y)) = A(A(x,y)^+,M(x,y))$$

Hence, by induction, for all x, $M(x,y) = M(y,x)$ holds for all y.

We next prove the distributivity of multiplication, and then, using that, we can prove that multiplication is associative.

We prove that $M(A(x,y),z) = A(M(x,z),M(y,z))$ holds for all x,y,z. The base case is when $z = 0$. This holds, as $A(M(x,0),M(y,0)) = A(0,0) = 0$ and $M(A(x,y),0) = 0$. Now suppose the result holds for some z, that is, $M(A(x,y),z) = A(M(x,z),M(y,z))$ holds for all x and y. Then

$$M(A(x,y),z^+) = A(A(x,y),M(A(x,y),z))$$
$$= A(A(x,y),A(M(x,z),M(y,z)))$$
$$= A(A(A(x,y),M(x,z)),M(y,z))$$
$$= A(A(A(y,x),M(x,z)),M(y,z))$$
$$= A(A(y,A(x,M(x,z))),M(y,z))$$
$$= A(A(A(x,M(x,z)),y),M(y,z))$$
$$= A(A(x,M(x,z)),A(y,M(y,z)))$$
$$= A(M(x,z^+),M(y,z^+))$$

therefore completing the proof.

Now we prove that multiplication is associateive. That is, we prove that $M(x,M(y,z)) = M(M(x,y),z)$ holds for all x,y,z. The base case is when $x = 0$. This holds, as $M(0,M(y,z)) = 0$ and $M(M(0,y),z) = M(0,z) = 0$. Now suppose the result holds for some x, that is, $M(x,M(y,z)) = M(M(x,y),z)$ holds for all y and z. Then

$$M(M(x^+,y),z) = M(M(y,x^+),z)$$

$$
\begin{aligned}
&= M(A(y, M(y, x)), z) \\
&= A(M(y, z), M(M(y, x), z)) \\
&= A(M(y, z), M(M(x, y), z)) \\
&= A(M(y, z), M(x, M(y, z)) \\
&= A(M(y, z), M(M(y, z), x)) \\
&= M(M(y, z), x^+) \\
&= M(x^+, M(y, z))
\end{aligned}
$$

therefore completing the proof.

Exercise 8.5

Are we sure that $\{a_0, a_1, ..., a_n, a_{n+1}, ...\}$ is a set? I cannot find anything that guarantees that.

Suppose that $a_0, a_1, ..., a_n, a_{n+1}, ...$ is an infinite sequence such that for each n, $a_n \subseteq a_{n+1}$. Let c be the class $\{a_0, a_1, ..., a_n, a_{n+1}, ...\}$. We prove that c is a nest, that is, for any two elements a and b of c, either $a \subseteq b$ or $b \subseteq a$. An obvious induction argument (one that has already been given in multiple other exercises) shows that, if $m \leq n$, then $a_m \subseteq a_n$. Now consider any two elements a and b of c. Thus $a = a_m$ and $b = a_n$ for some m and n. If $m = n$, the both $a \subseteq b$ and $b \subseteq a$ hold. If $m < n$, then $a \subseteq b$ holds. If $n < m$, then $b \subseteq a$ holds. Thus the proof is complete.

Exercise 8.6

Again, I am not sure why $\{a_0, a_1, ..., a_n, a_{n+1}, ...\}$ is guaranteed to be a set in this exercise.

Suppose that g is a function and $a_0, a_1, ..., a_n, a_{n+1}, ...$ is an infinite sequence such that $a_0 = \emptyset$, and for each n, $a_{n+1} = g(a_n)$. Let c be the class $\{a_0, a_1, ..., a_n, a_{n+1}, ...\}$. We prove that c is the class which is minimally inductive under g. Obviously, c is inductive under g. Now consider any proper subclass S of c. The class of all elements in c but not in S must have an element with a least index n. Either $a_n = \emptyset$ or $a_n = g(a_{n-1})$. In the first case, S would not contain $\emptyset$, meaning it would not be g-inductive. In the second case, because n is the least index of all elements in c but not in S, $a_{n-1} \in S$. Thus $a_{n-1} \in S$, but $g(a_{n-1}) \notin S$, meaning S would not be g-inductive. Thus, in any case, S is not g-inductive. Hence no proper subclass of c is g-inductive, meaning c is minimally g-inductive.

Section 9

Exercise 9.1(a)

Suppose that M is minimally inductive under g and that R is a relation satisfying the following two conditions:
D_1' $R(x, 0)$ and $R(0, x)$ hold for every x in M.
D_2' For every x and y in M, $[R(x, y) \wedge R(x, g(y)) \wedge R(g(x), y)] \supset R(g(x), g(y))$.
We prove that $R(x, y)$ holds for all x, y in M. As in the first double induction principle, we call an element x of M left-normal if $R(x, y)$ holds for all y in M. We first show that every element of M is left normal. Now, 0, is left normal, as it was assumed that $R(0, x)$ holds for all x. Now suppose that some x in M is left normal. We need to show that $R(g(x), y)$ holds for all y. We are given that $R(g(x), 0)$ holds. Now suppose that $R(g(x), y)$ holds for some y. Because x is left normal, we know that $R(x, g(y))$ and $R(x, y)$ both hold. Thus $R(x, y)$, $R(g(x), y)$, and $R(x, g(y))$ all hold, meaning, by D_2', $R(g(x), g(y))$ holds. Thus, by g-induction, $R(g(x), y)$ holds for all y, meaning $g(x)$ is left normal. Thus, by g-induction, all elements of M are left normal. Now consider any elements x, y of M. Then, because x is left normal, $R(x, y)$ holds, therefore completing the proof.

Exercise 9.1(b)

We suppose that g is slowly progressing and $R(x, y)$ is the relation $x \subseteq y$ or $y \subseteq x$. We prove that R satisfies conditions D_1', D_2' of the Theorem S_1. For this, we first prove the following Lemma:
Lemma:*If g is slowly progressing, then if $x \subseteq y \subseteq g(x)$, either $y = x$ or $y = g(x)$*
We suppose that $x \subseteq y \subseteq g(x)$. Now, because g is slowly progressing, either $x = g(x)$ or $g(x)$ contains exactly one more element than x. In the first case, obviously $y = x = g(x)$. In the second case, suppose that $x \neq y$. Then $x \subset y$. Now, consider the element e in $g(x)$ but not in x. Because it is the only such element in $g(x)$ but not in x, and because $y \subseteq g(x)$, $e \in y$. And there could be no other such element in y but not in x; for then y would not be a subset of $g(x)$. Hence $y = g(x)$, thus completing the proof.

Having proven the above Lemma, Lemma S_2 follows easily. For suppose that g is slowly progressing and that $R(x, y)$ is the relation $x \subseteq y$ or $y \subseteq x$. D_1' obviously holds (as 0 is just the empty set). Now suppose that $R(x, y) \wedge R(x, g(y)) \wedge R(g(x), y)$ held. On the one hand, suppose that $x \subseteq y$. Then there are two sub-cases: either $g(x) \subseteq y$, or $y \subseteq g(x)$. The first obviously means that $g(x) \subseteq g(y)$, meaning $R(g(x), g(y))$. On the other hand, suppose that $y \subseteq g(x)$. Then we know that $x \subseteq y \subseteq g(x)$, meaning, by the Lemma, either $y = x$ or $y = g(x)$. In the first case, $g(y) = g(x)$ and obviously $R(g(x), g(y))$. In the second case, $g(x) \subseteq g(g(x)) = g(y)$, meaning $R(g(x), g(y))$ holds. Thus, in any case, $R(g(x), g(y))$ holds. On the other hand, if we had $y \subseteq x$, the proof is nearly identical. Thus the proof is complete.

Exericse 9.1(c)

(1) Suppose that $x \subset y \subset g(x)$ held. Then $x \subseteq y \subseteq g(x)$, meaning, by the Lemma in Exercise 9.1(b), either $x = y$ or $y = g(x)$, which would mean that either $x \subset y$ or $y \subset g(x)$ did not hold, a contradiction. Thus $x \subset y \subset g(x)$ can never hold. (This part of the proof is unnecessary, considering we have already proven the Lemma in exercise 9.1(b). Perhaps Smullyan's original proof did not first prove that Lemma. Indeed, $x \subset y \subset g(x)$ never holding implies that $x \subseteq y \subseteq g(x)$ means either $x = y$ or $y = g(x)$).

(2) Suppose that N is a nest closed under g (and that g is slowly progressing). For any elements x and y of N, we know that either $x \subseteq y$ or $y \subseteq x$. So suppose $x \subseteq y$. We also know that either $g(x) \subseteq y$ or $y \subseteq g(x)$. Suppose that $y \subseteq g(x)$. Then, by the Lemma in Exercise 9.1(b), either $x = y$ or $y = g(x)$. In the first case, obviously $y \subseteq x$. In the second case, obviously $g(x) \subseteq y$. Thus the proof is complete.